



Optical Properties 2

Presented by Dr Adijat 'Wumi Inyang

Optical properties

- **Scattering of light**

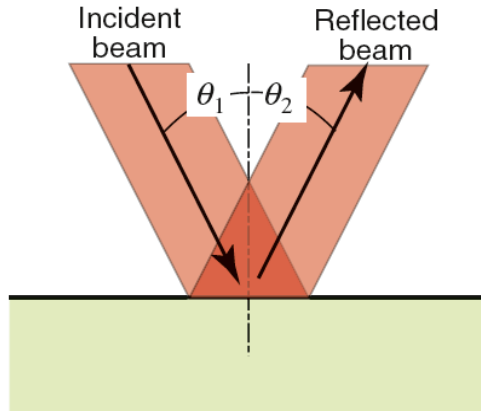
- Atoms and molecules absorb and reflect light in different directions
- Light is forced to change its pathway due to interaction with atoms, molecules and particles
- The scattering depends on the composition of the material, impurities, crystal structure, size of molecules and surface roughness
- Reflection of radiation due to scattering are included in this category



Optical properties

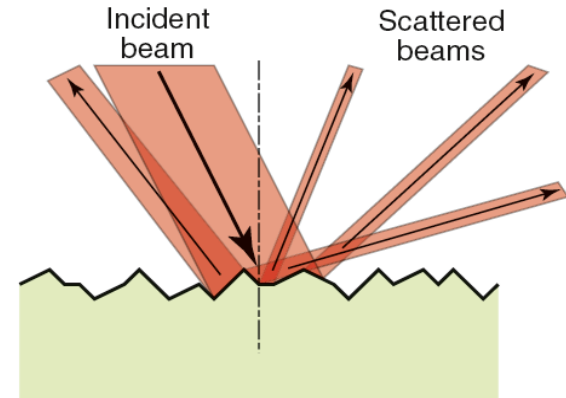
- **Reflection**

Specular Reflection



Specular surfaces are smooth and flat, reflecting light so that $\theta_1 = \theta_2$

Diffuse Reflection



Diffuse surfaces are irregular, causing light to scatter when reflected

Optical properties

- **Scattering of light in polymers**

- For highly **amorphous** and pore-free polymers
 - Little or no scattering
 - These materials are transparent
- **Semicrystalline** polymers
 - Different indices of refraction for amorphous and crystalline regions
 - Scattering of light at boundaries
 - Highly crystalline polymers may be opaque

- **Examples:**

- Polystyrene (amorphous)
clear and transparent
- Low-density polyethylene
milk cartons – opaque



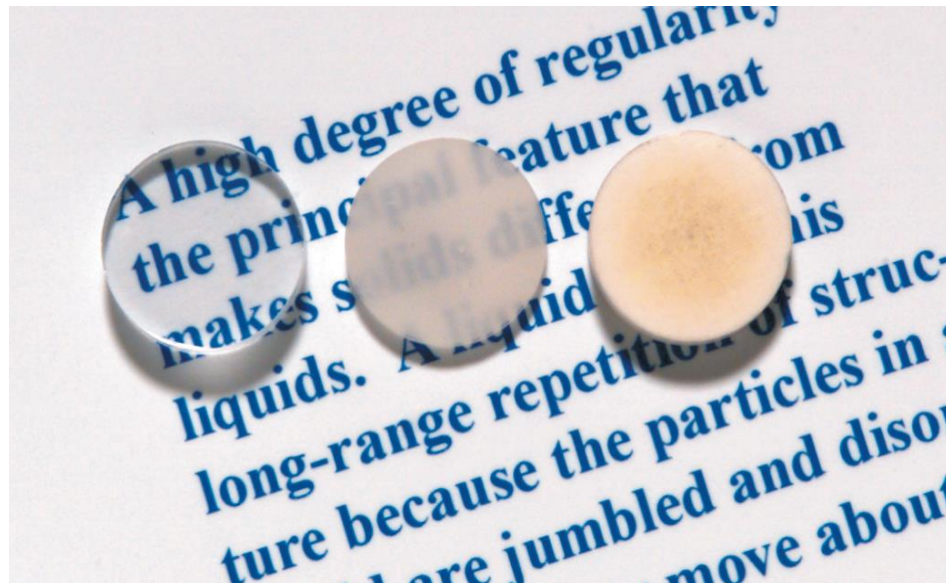
Polystyrene (amorphous)
– clear and transparent



Low-density polyethylene
– opaque

Optical properties

- Scattering of light in ceramics



single
crystal

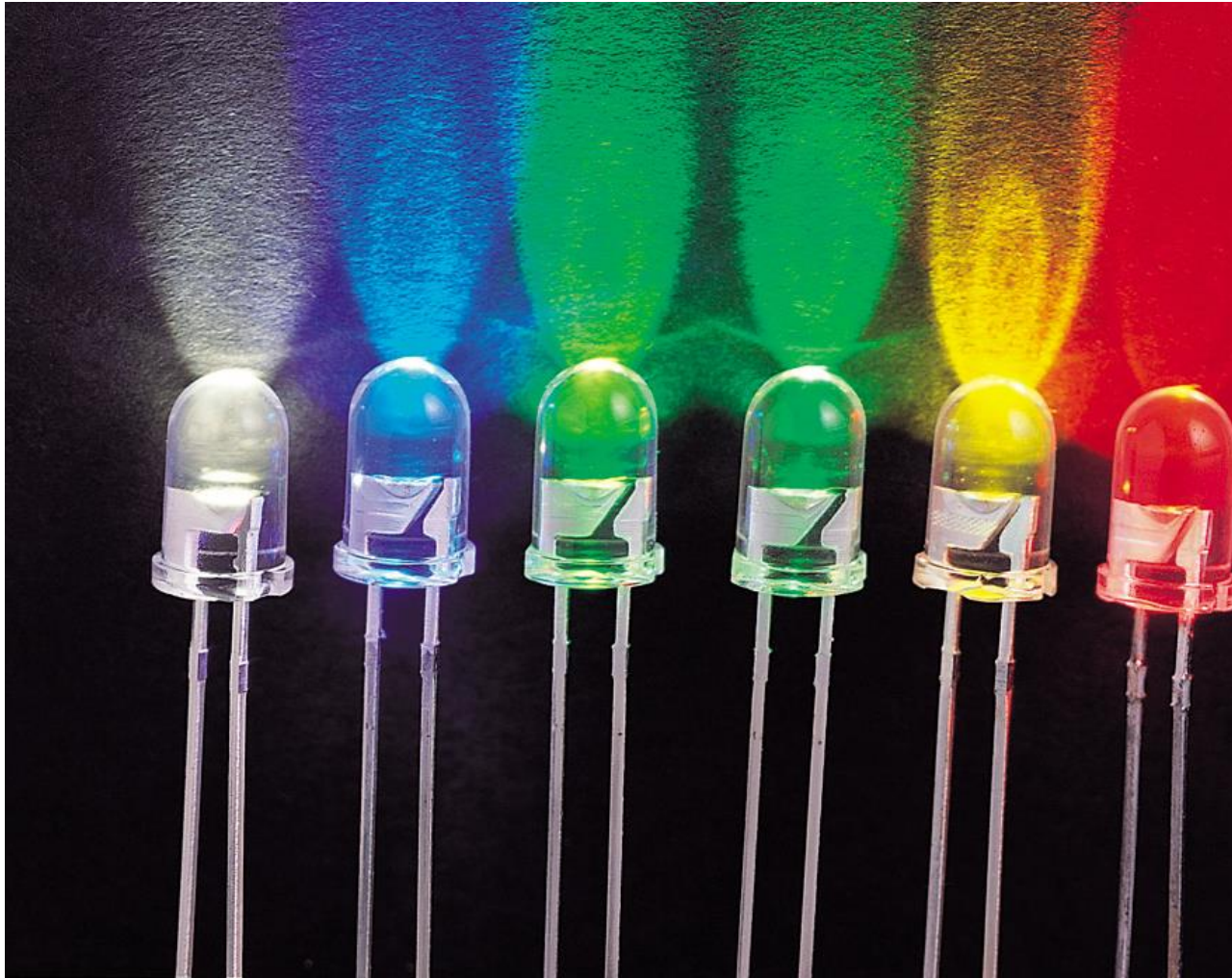
polycrystalline
dense

polycrystalline
porous

Example for Al_2O_3 :

- Single crystal alumina (sapphire) is transparent
- In polycrystalline alumina – light scatters from the grain boundaries
- If the alumina is not fully dense (porous) light scatters from the surface of pores

Optical technology



Light emitting diode (LED)

- **Energy efficient white light LEDs – Nobel Prize in Physics 2014**
 - Isamu Akasaki, Hiroshi Amano and Shuji Nakamura produced bright blue light beams from semi-conductors in the early 1990s
 - The blue LEDs are the result of an adequate crystal composition and structure of Gallium Nitride (GaN)
 - The band gap for these semiconductors is low enough for an electron transition to occur to the conductive band
 - The transition allows good optical properties of generating white light



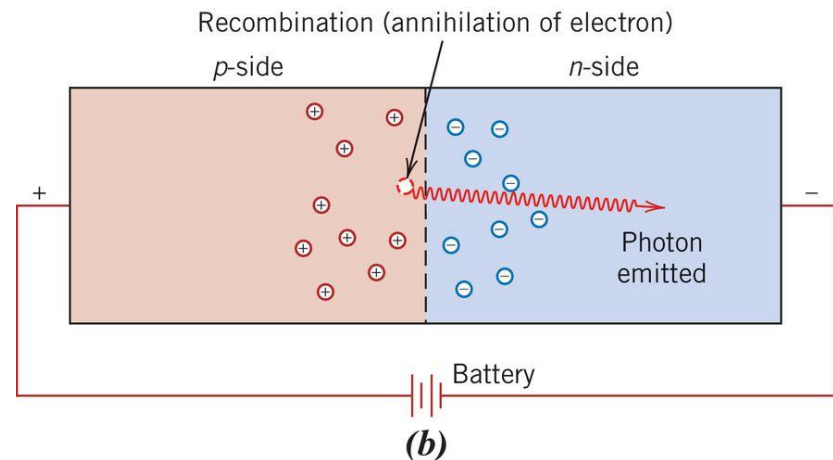
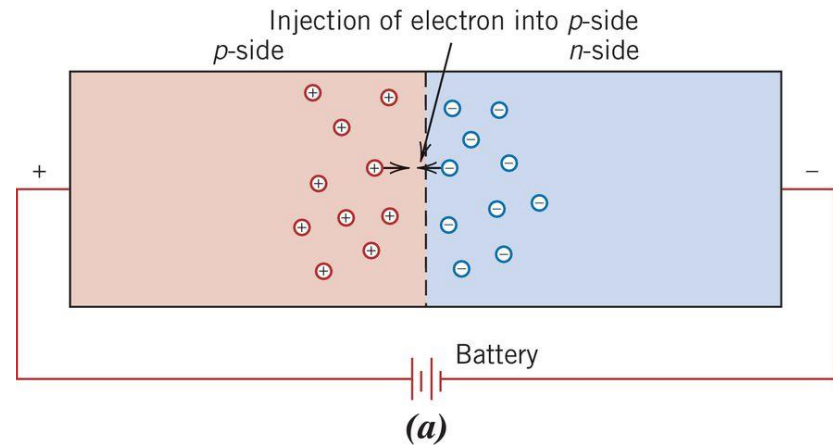
Light emitting diode (LED)

- p-n junction diode**

In forward bias: carriers flow through p -type and n -type regions; holes and electrons recombine at p - n junction;

The recombination is the transition of the conduction electron across the band gap to the hole in the valence band

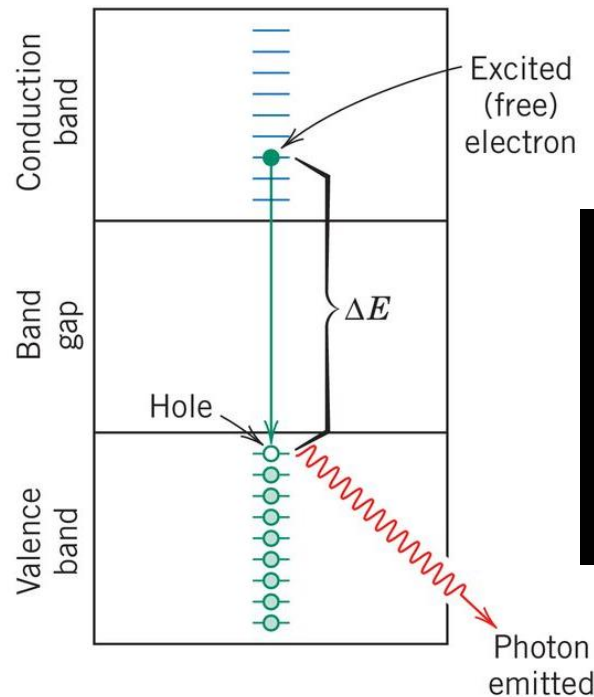
The difference in energy is emitted as a photon
– the wavelength (colour) depending on the band gap



Light emitting diode (LED)

- electron – hole recombination

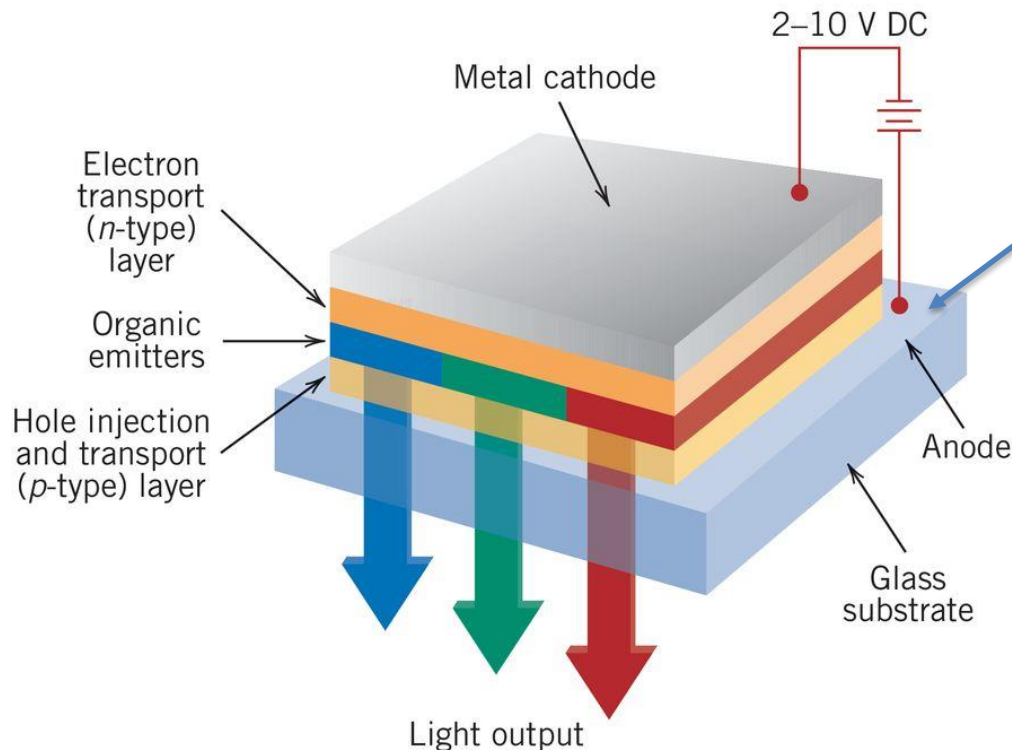
Material	Band Gap (eV)
Ge	0.67
Si	1.11
AIP	2.42
AlSb	1.58
GaAs	1.42
GaP	2.26
InP	1.35
InSb	0.17
CdS	2.40
CdTe	1.56
ZnS	3.66
ZnTe	2.4



The difference in energy is emitted as a photon
the wavelength (colour) depending on the band gap

Light emitting diode (LED)

- **organic LEDs**



Transparent conductor
For anode material
Indium Tin Oxide (ITO)
In74% Sn18% O8%

Polymer LEDs can be
made to emit light of
different colours

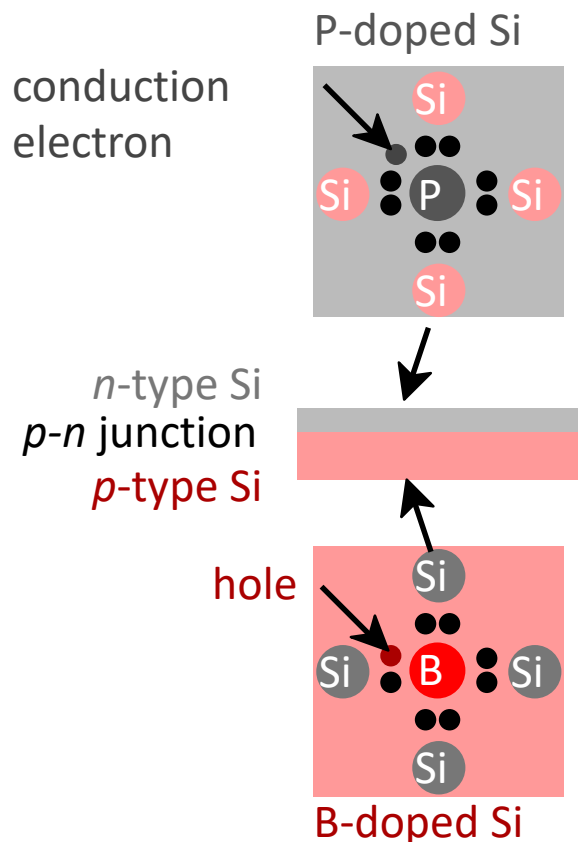
Polymers are low energy
to make and can be
printed to make displays

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Light absorption: semi-conductors

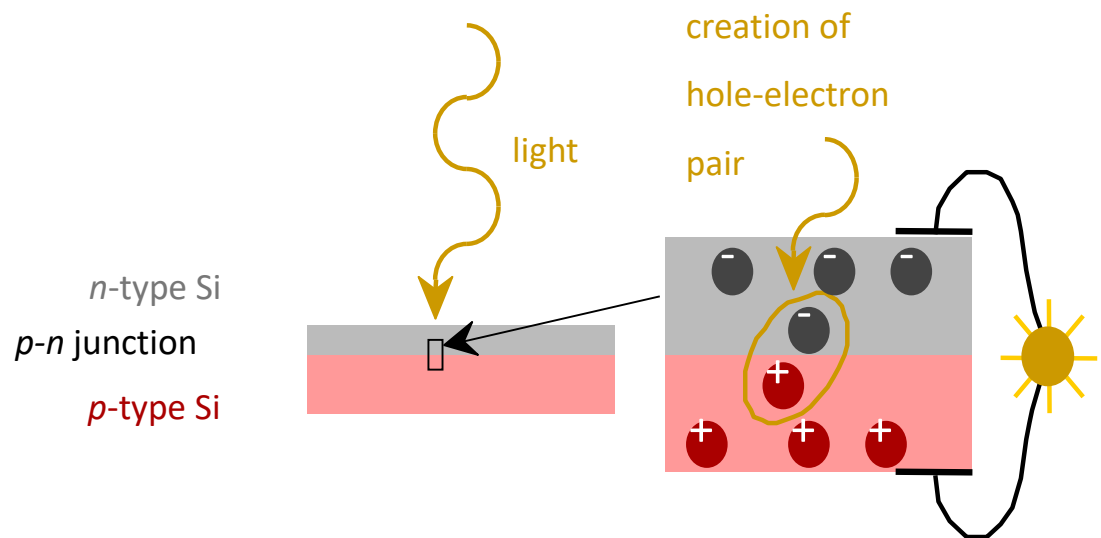
- Optical absorption photo-voltaic (pv) **solar panels**

- *p-n* junction:



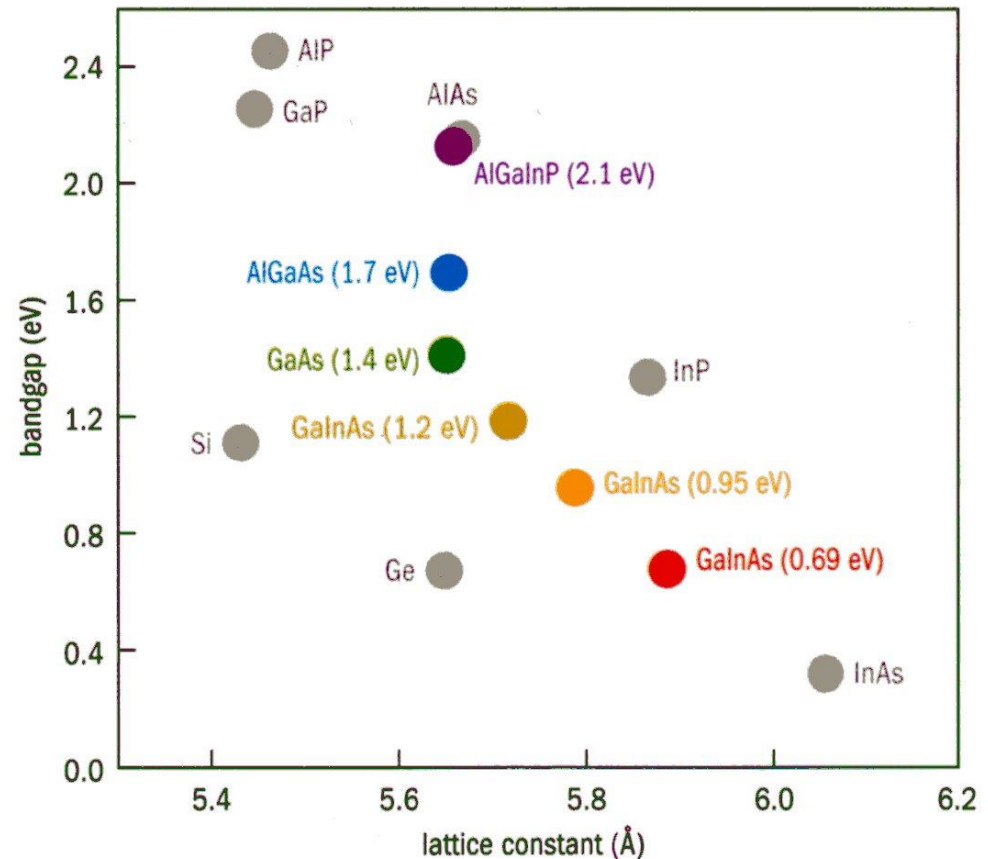
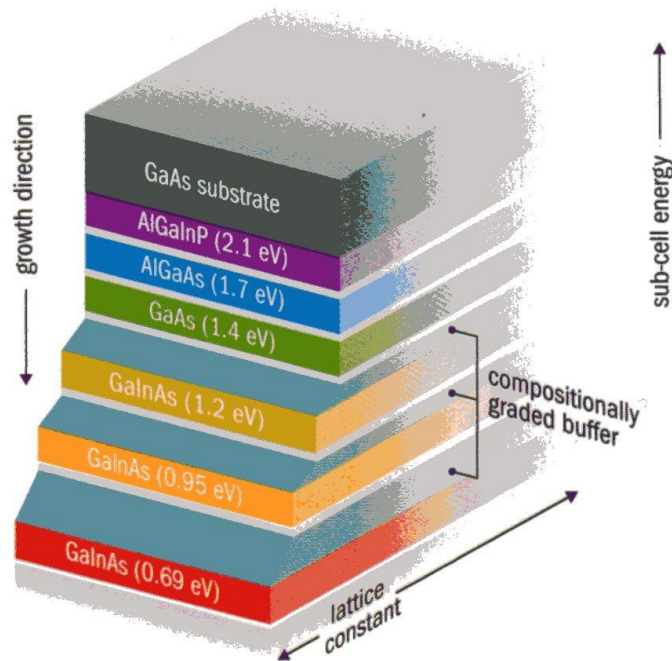
- Operation:

- incident photon of light produces electron-hole pair.
- typical potential of 0.5 V produced across junction
- current increases w/light intensity.



Light absorption: semi-conductors

- Optical absorption from semiconductors
band gap engineering



Physics World: April 2021

By controlling the composition and internal strain of the semi-conductor layer, the band gap can be 'engineered' to absorb any colour in the visible spectrum
Used to make full-spectrum photo-voltaic (pv) solar panels

Lasers



- The laser generates light waves that are in phase (coherent) and that travel parallel to one another
 - **LASER**
 - Light
 - Amplification by
 - Stimulated
 - Emission of
 - Radiation
- Operation of laser involves a **population inversion** of energy states process
- More electrons in excited states than the ground state

Lasers

■ Stimulated Emission

- The generation of one photon by the decay transition of an electron, induces the emission of other photons that are all in phase with one another.
 - This cascading effect produces an intense burst of coherent light.
-
- This is an example of a **pulsed laser**

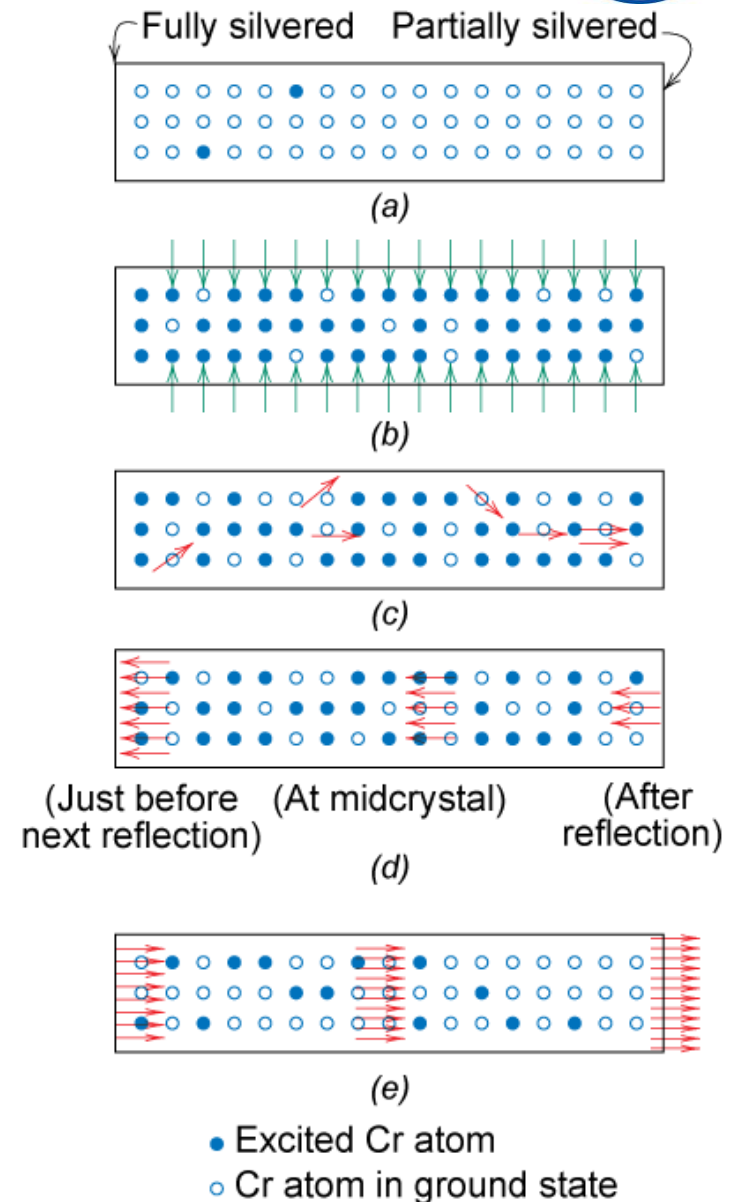


Fig. 22.15, Callister & Rethwisch 9e.

Lasers

• Solid state semi-conductor lasers

- Apply strong forward bias across semiconductor layers, metal, and heat sink.
- Electron-hole pairs generated by electrons that are excited across band gap.
- Recombination of an electron-hole pair generates a photon of laser light

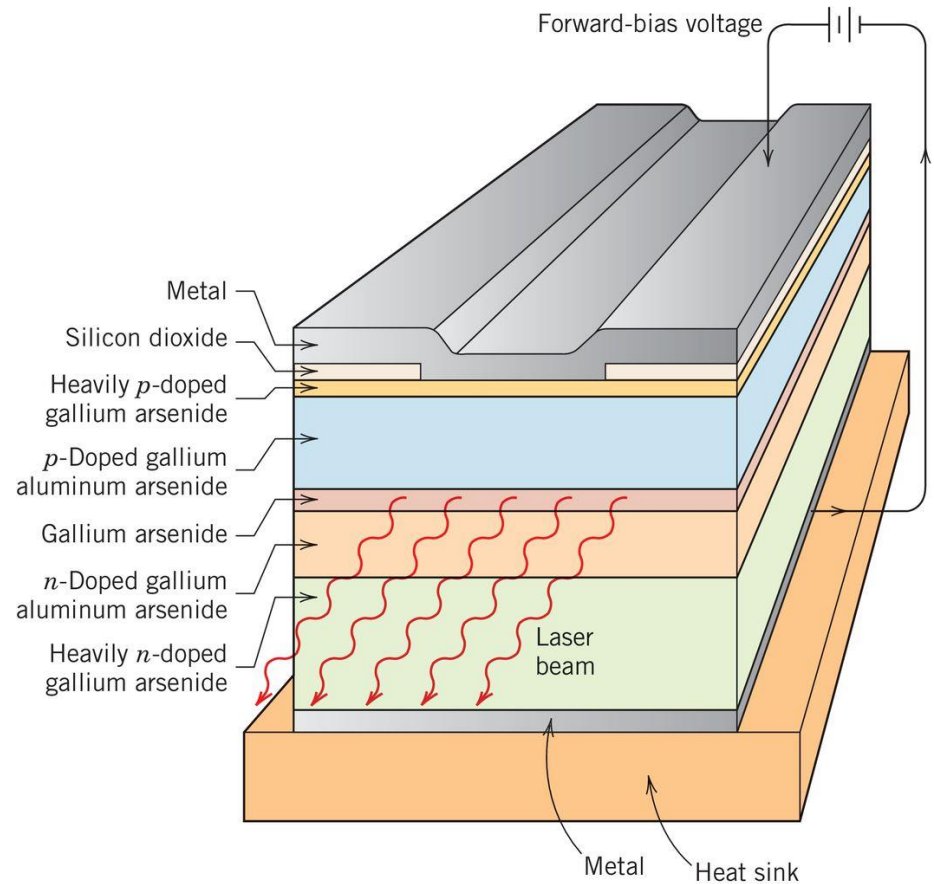
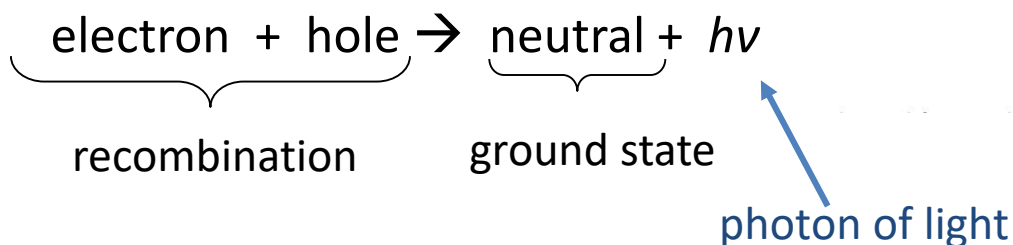


Fig. 22.17, Callister & Rethwisch 9e.

Characteristics of lasers

<i>Laser</i>	<i>Wavelength (μm)</i>	<i>Average Power Range</i>	<i>Applications</i>
Carbon dioxide	10.6	Milliwatts to tens of kilowatts	Heat treating, welding, cutting, scribing, marking
Nd:YAG	1.06	Milliwatts to hundreds of watts	Welding, hole piercing, cutting
	0.532	Milliwatts to watts	
Nd:glass	1.05	Watts ^a	Pulse welding, hole piercing
Diodes	Visible and infrared	Milliwatts to kilowatts	Bar-code reading, CDs and DVDs, optical communications
Argon-ion	0.5415	Milliwatts to tens of watts	Surgery, distance measurements, holography
	0.488	Milliwatts to watts	
Fiber	Infrared	Watts to kilowatts	Telecommunications, spectroscopy, directed energy weapons
Excimer	Ultraviolet	Watts to hundreds of watts ^b	Eye surgery, micromachining, microlithography

^aAlthough glass lasers produce relatively low average powers, they almost always run in pulsed mode, where their peak powers can reach the gigawatt levels.

^bExcimers are also pulsed lasers, and are capable of peak powers in the tens of megawatts.

Optical fibres

- **Metals are opaque materials that can absorb light**
- Fibers have diameters of $125\mu\text{m}$ or less
- Made of high purity silica glass, they are flaw free and strong
- Plastic cladding $60\mu\text{m}$ thick is applied to fibers
- Containment of the light within the fiber core is possible by **total internal reflection**
- The core is clad with a lower refractive index material

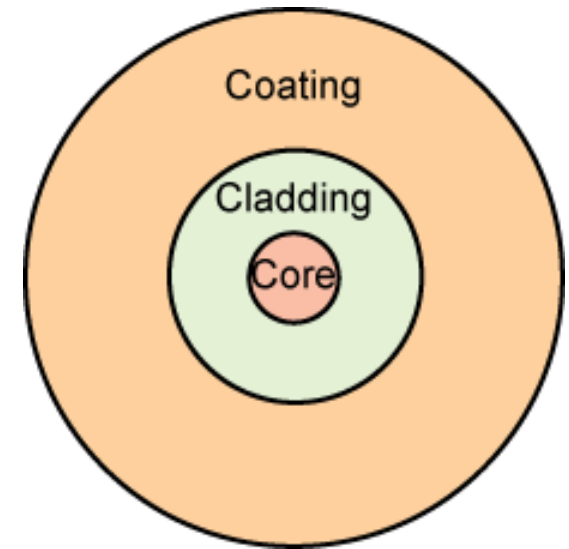
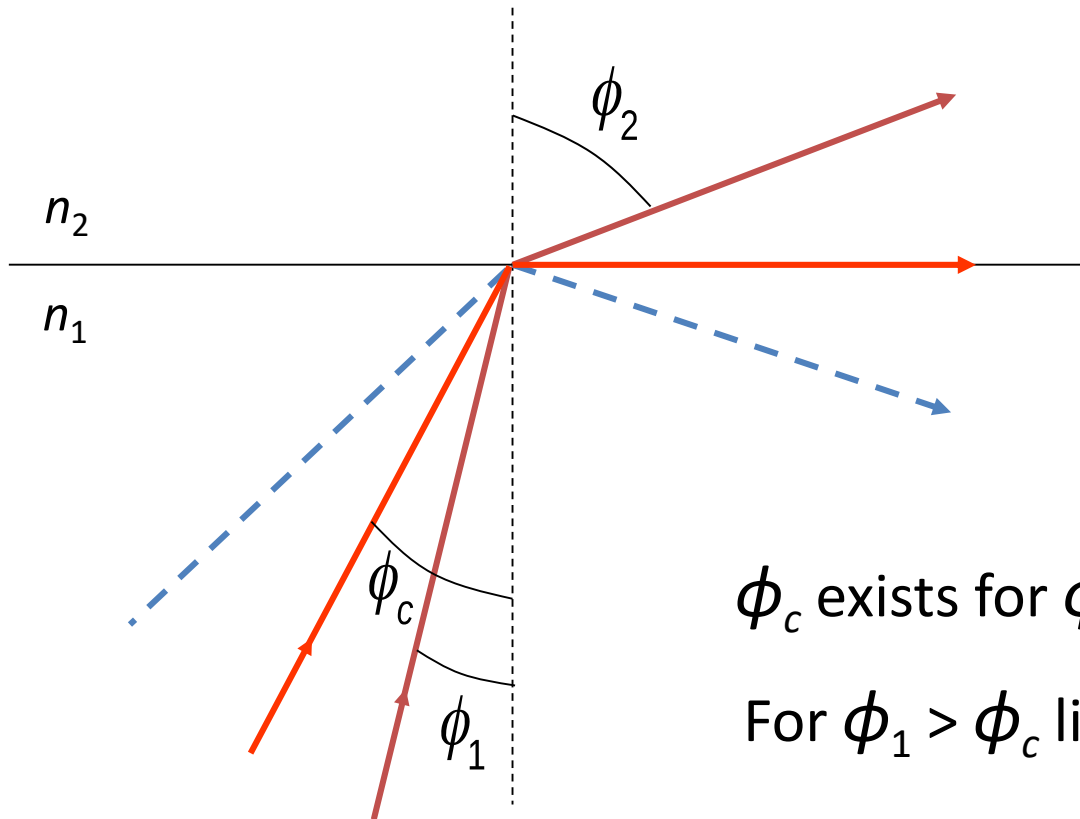


Fig. 22.20, Callister & Rethwisch 9e.

Total internal reflectance



$$\frac{n_1}{n_2} = \frac{\sin \phi_2}{\sin \phi_1}$$

ϕ_1 = incident angle

ϕ_2 = refracted angle

ϕ_c = critical angle

ϕ_c exists for $\phi_2 = 90^\circ$

For $\phi_1 > \phi_c$ light is internally reflected

$$\sin \phi_c = \frac{n_2}{n_1} \quad n_2 < n_1$$

Optical fibres

Step-index Optical Fiber

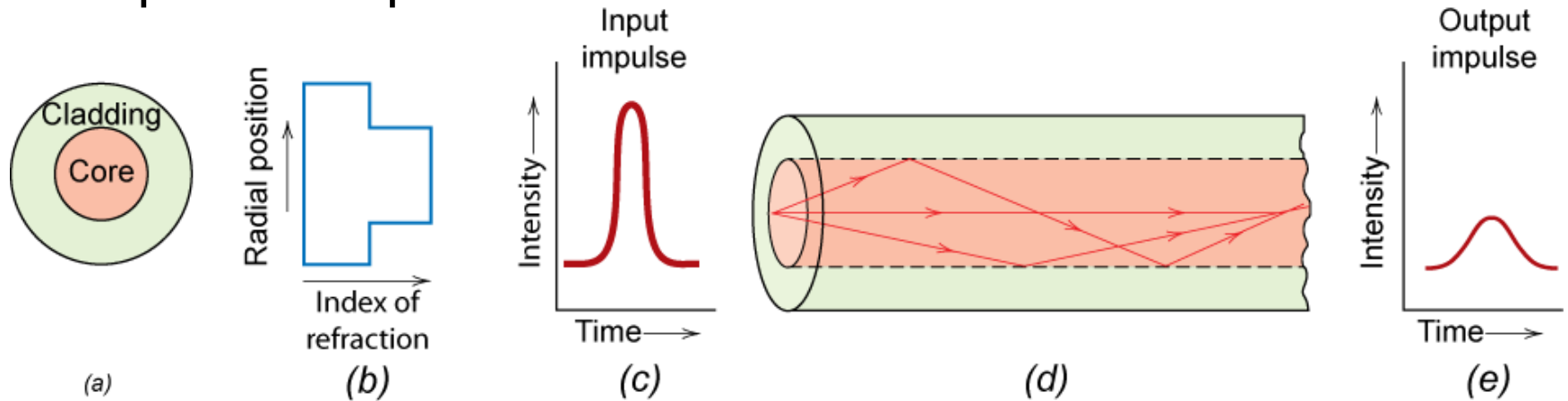


Fig. 21.21, Callister & Rethwisch 9e.

Graded-index Optical Fiber

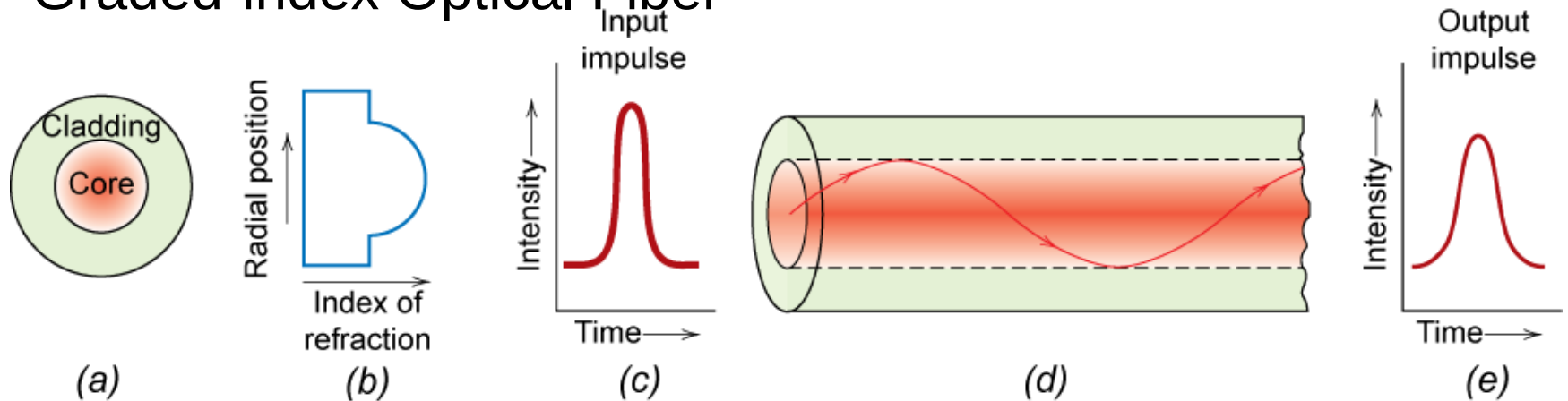
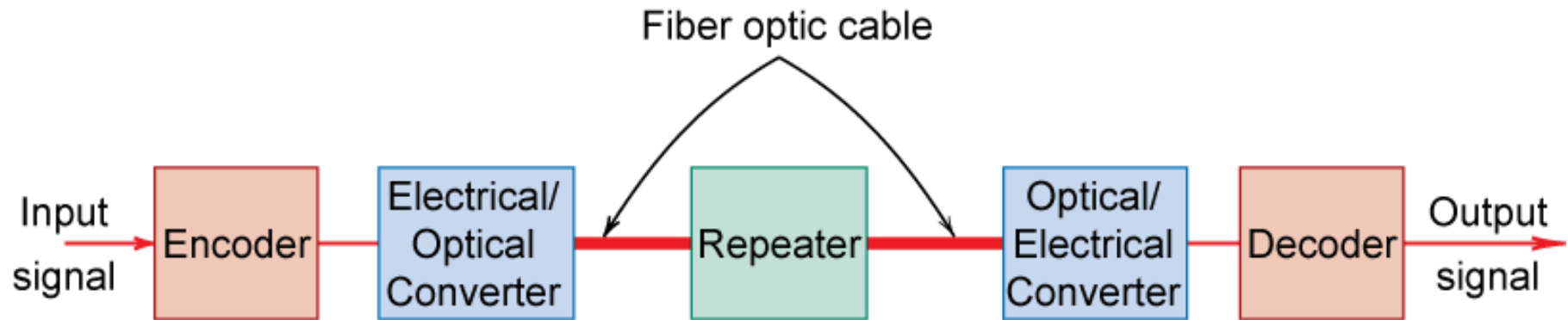


Fig. 21.22, Callister & Rethwisch 9e.

Optical communications

Digital communications by optical fibre are much more efficient than electric communications by copper wire

Schematic diagram showing components of a fiber optic communications system



A single optical fibre can carry 250 million phone conversations every second. It would take 30 tonnes of copper to transmit the same information.

Optical properties:

- Scattering of light from and in materials
- **Optical technologies**
 - LEDs
 - Photo-voltaics (pv) solar panels
 - Lasers
 - Optical fibres